

DIGITAL CLOCK

Inputs

Port	Type	Width	MSB/LSB
clk	Input	1 bit	clk itself
reset	Input	1 bit	reset itself
enable	Input	1 bit	enable itself

So these are single-bit inputs, not buses.

Outputs

Port	Type	Width	MSB	LSB
hours	Output	5-bit bus	hours[4]	hours[0]
minutes	Output	6-bit bus	minutes[5]	minutes[0]
seconds	Output	6-bit bus	seconds[5]	seconds[0]

So the port declaration is:

↓

Main Code

```
module digital_clock #(
    parameter CLK_FREQ = 100_000_000
)(
    input wire    clk,
    input wire    reset,
    input wire    enable,

    output reg [4:0] hours,
    output reg [5:0] minutes,
    output reg [5:0] seconds
);

// Counter for generating 1-second pulse
```

```
reg [26:0] count;
```

```
always @(posedge clk or posedge reset)
```

```
begin
```

```
    if (reset)
```

```
    begin
```

```
        count <= 0;
```

```
        hours <= 0;
```

```
        minutes <= 0;
```

```
        seconds <= 0;
```

```
    end
```

```
    else if (enable)
```

```
    begin
```

```
        // Generate 1-second timing
```

```
        if (count == CLK_FREQ - 1)
```

```
        begin
```

```
            count <= 0;
```

```
        // Increment seconds
```

```
        if (seconds == 59)
```

```
        begin
```

```
            seconds <= 0;
```

```
        // Increment minutes
```

```
        if (minutes == 59)
```

```
        begin
```

```
            minutes <= 0;
```

```
        // Increment hours
```

```
        if (hours == 23)
```

```
        begin
            hours <= 0;
        end
    else
        begin
            hours <= hours + 1;
        end
    end
else
    begin
        minutes <= minutes + 1;
    end
end
else
    begin
        seconds <= seconds + 1;
    end
end
else
    begin
        count <= count + 1;
    end
end
end
end

endmodule
```

Test Bench Code

```
`timescale 1ns/1ps
```

```

module digital_clock_tb;

    reg clk;
    reg reset;
    reg enable;

    wire [4:0] hours;
    wire [5:0] minutes;
    wire [5:0] seconds;

    // Instantiate Digital Clock
    digital_clock #(
        .CLK_FREQ(10)
    )
    uut (
        .clk(clk),
        .reset(reset),
        .enable(enable),
        .hours(hours),
        .minutes(minutes),
        .seconds(seconds)
    );

    // Clock generation
    always #5 clk = ~clk;

    initial
    begin
        // Initial values
        clk = 0;
        reset = 1;
    end

```

```
enable = 0;

// Reset
#20;
reset = 0;

// Start clock
enable = 1;

// Run simulation
#700;

// Stop clock
enable = 0;

#50;

$finish;
end

// Display values
initial
begin
    $monitor("Time = %02d:%02d:%02d",
             hours, minutes, seconds);
end

endmodule
```